ipv unit 3

Unit-3: Object Motion and Tracking (Computer Vision)

RGPV Examination Notes (Notion Format)

Question 1. Explain Motion Detection in Digital Video Processing.

Definition

Motion Detection is the process of identifying moving objects in a sequence of video frames by comparing consecutive frames.

It is the first step in object tracking and motion analysis.

Examples

- CCTV Surveillance
- Traffic Monitoring
- Human Activity Recognition
- Self Driving Cars

Basic Principle

Motion is detected by calculating the difference between two consecutive frames.

Formula

[
D(x,y)=|I*t(x,y)-I*{t-1}(x,y)|
]

Where:

- (I_t) = Current Frame
- (I_{t-1}) = Previous Frame
- (D(x,y)) = Difference Image

If

$$
D(x,y)>Threshold
$$

Motion exists.

Flowchart

```
Video Input
      ↓
Capture Consecutive Frames
      ↓
Frame Differencing
      ↓
Thresholding
      ↓
Motion Region Detection
      ↓
Object Tracking
```

Advantages

Simple

Easy implementation.

Fast

Suitable for real-time applications.

Low Cost

Requires less computation.

Disadvantages

Noise Sensitive

Noise may be detected as motion.

Illumination Sensitive

Lighting changes affect detection.

Slow Object Problem

Very slow objects may not be detected.

Applications

- Surveillance
- Robotics
- Traffic Analysis
- Gesture Recognition

RGPV Question

Q. Explain Motion Detection with neat diagram.

Q. Describe Frame Differencing Method for Motion Detection.

Question 2. Explain Motion Estimation Models.

Definition

Motion Estimation determines the displacement and direction of moving objects between frames.

Types of Motion Models

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1. Global Motion Model

Entire image moves together.

Causes

- Camera Rotation
- Camera Translation
- Zooming

Example

Moving a mobile camera.

Diagram

```
Frame 1                Frame 2

Entire Scene  →  Entire Scene Shifted
```

Advantages

- Easy representation
- Useful for camera motion compensation

Disadvantages

- Cannot track individual objects

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2. Local Motion Model

Only selected objects move.

Example

Moving car in a static road scene.

Diagram

```
Road (Static)

Car →→→ Moving
```

Advantages

- Accurate object tracking
- Useful in surveillance

Disadvantages

- Computationally expensive

Comparison Table

Feature	Global Motion	Local Motion
Movement	Entire Frame	Specific Objects
Complexity	Low	High
Accuracy	Low	High
Usage	Camera Motion	Object Tracking

RGPV Questions

Q. Differentiate between Global and Local Motion Models.

Q. Explain Motion Estimation Models with examples.

Question 3. Explain Block Matching Algorithm (BMA).

Definition

Block Matching is a technique used to estimate motion by dividing an image into blocks and searching for matching blocks in another frame.

Working Steps

Step 1

Divide image into blocks.

Step 2

Select current block.

Step 3

Search matching block.

Step 4

Compute displacement vector.

Flowchart

```
Input Frames
      ↓
Divide into Blocks
      ↓
Search Matching Block
      ↓
Calculate Motion Vector
      ↓
Motion Estimation
```

Motion Vector

[
MV=(dx,dy)
]

Where

dx = Horizontal displacement

dy = Vertical displacement

Numerical

Suppose a block center moves:

Frame 1

[
(100,150)
]

Frame 2

[
(108,145)
]

Find motion vector.

Solution

[
dx=108-100=8
]

[
dy=145-150=-5
]

Motion Vector

[
MV=(8,-5)

]

Answer:

8 pixels right and 5 pixels upward.

Advantages

- Easy implementation
- Widely used in video compression

Disadvantages

- High search cost
- Less accurate for complex motion

RGPV Questions

Q. Explain Block Matching Algorithm.

Q. Explain Motion Vector with suitable example.

Numerical:

Find motion vector if object position changes from (50,75) to (60,90).

Question 4. Explain Optical Flow.

Definition

Optical Flow estimates apparent motion of pixels between consecutive frames.

It calculates how brightness patterns move over time.

Concept

If an object moves, its pixels also move.

Optical Flow tracks pixel displacement.

Diagram

```
Frame 1

● ● ●

Frame 2

    ● ● ●

Pixel Motion →
```

Optical Flow Constraint Equation

[
I_xu + I_yv + I_t=0
]

Where

- (I_x) = Intensity change in x direction
- (I_y) = Intensity change in y direction
- (I_t) = Intensity change with time
- (u,v) = Velocity components

Types

Sparse Optical Flow

Tracks selected points.

Dense Optical Flow

Tracks all pixels.

Advantages

High Accuracy

Pixel-level motion estimation.

No Prior Training

Works directly on images.

Disadvantages

Computationally Expensive

Dense flow requires more processing.

Sensitive to Lighting

Illumination changes affect results.

Applications

- Self Driving Cars
- Video Surveillance
- Robotics
- Medical Imaging

RGPV Questions

Q. Explain Optical Flow with mathematical formulation.

Q. What are Sparse and Dense Optical Flow?

Q. Explain Optical Flow Constraint Equation.

Question 5. Explain Lucas-Kanade Optical Flow Method.

Definition

Lucas-Kanade is a local optical flow algorithm that assumes neighboring pixels move similarly.

Assumptions

Brightness Constancy

Brightness remains unchanged.

Small Motion

Motion between frames is small.

Neighboring Pixels Move Together

Local region has same motion.

Algorithm

```
Input Frames
      ↓
Select Feature Window
      ↓
Compute Gradients
      ↓
Solve Optical Flow Equation
      ↓
Motion Vector
```

Advantages

- Fast

- Efficient
- Real Time Performance

Disadvantages

- Cannot handle large motion
- Sensitive to noise

RGPV Questions

Q. Explain Lucas-Kanade Optical Flow Method.

Q. State assumptions of Lucas-Kanade Algorithm.

Question 6. Explain KLT Tracker.

(Kanade-Lucas-Tomasi Tracker)

Definition

KLT Tracker tracks good feature points using Lucas-Kanade Optical Flow.

Working

Step 1

Detect corners.

Step 2

Select strong corners.

Step 3

Track corners across frames.

Flowchart

```
Video Frames
      ↓
Corner Detection
      ↓
Feature Selection
      ↓
Lucas Kanade Tracking
      ↓
Object Trajectory
```

Advantages

Accurate

Tracks strong features.

Robust

Works under partial occlusion.

Disadvantages

Feature Drift

Errors accumulate.

Large Motion Failure

Tracking may fail.

RGPV Questions

Q. Explain KLT Tracker with diagram.

Q. Compare Lucas-Kanade and KLT Tracker.

Question 7. Explain Feature Detection and Tracking.

Definition

Feature Tracking follows important image points across multiple frames.

Common Features

Corners

Edges

Blobs

Feature Detection Methods

Harris Corner Detector

Detects corners using intensity variations.

Advantages

- Fast
- Rotation Invariant

Disadvantages

- Scale Sensitive

SIFT

Scale Invariant Feature Transform

Features

- Scale Invariant
- Rotation Invariant

Advantages

- Highly Accurate

Disadvantages

- Computationally Expensive

ORB

Oriented FAST and Rotated BRIEF

Advantages

- Fast
- Lightweight

Disadvantages

- Less accurate than SIFT

Comparison Table

Feature	Harris	SIFT	ORB
Speed	Fast	Slow	Very Fast
Accuracy	Medium	High	Medium
Scale Invariant	No	Yes	Partial
Real Time	Yes	No	Yes

RGPV Questions

Q. Explain Harris Corner Detector.

Q. Compare SIFT and ORB.

Q. Explain Feature Tracking.

Question 8. Explain Feature Matching and RANSAC.

Feature Matching

Finding corresponding points between two images.

Process

```
Image 1
    ↓
Feature Detection
    ↓
Descriptor Generation
    ↓
Feature Matching
    ↓
Image Alignment
```

Similarity Measures

Euclidean Distance

Used in SIFT

[
D=\sqrt{\sum(x_i-y_i)^2}

]

Hamming Distance

Used in ORB

Counts different bits.

Numerical

Descriptor A

```
10101010
```

Descriptor B

```
10111010
```

Difference = 1 bit

Hamming Distance = 1

RANSAC

Random Sample Consensus

Used to remove incorrect feature matches.

Steps

```
Feature Matches
      ↓
Random Sampling
      ↓
Model Estimation
      ↓
```

```
Outlier Removal
      ↓
Best Match Set
```

Advantages

- Removes noise
- Improves matching accuracy

Disadvantages

- Computationally expensive
- Results depend on iterations

RGPV Questions

Q. Explain RANSAC Algorithm.

Q. Explain Feature Matching.

Numerical:

Calculate Hamming Distance between

11010101

11011101

Question 9. Explain Video Mosaicking.

Definition

Video Mosaicking combines multiple images or frames into a single panoramic image.

Steps

```
Image Acquisition
      ↓
Feature Detection
      ↓
Feature Matching
      ↓
RANSAC
      ↓
Image Warping
      ↓
Panorama Generation
```

Advantages

- Large field of view
- Better visualization

Applications

- Satellite Imaging
- Medical Imaging
- Google Street View

RGPV Questions

Q. Explain Video Mosaicking with block diagram.

Q. Explain Image Stitching Process.

Most Important 14-Mark Questions

1. Explain Optical Flow with mathematical formulation and applications.
2. Explain Lucas-Kanade Optical Flow Method.
3. Explain Feature Detection and Tracking Techniques.
4. Compare Harris, SIFT and ORB.
5. Explain RANSAC Algorithm and Feature Matching.
6. Explain Video Mosaicking with neat diagram.
7. Explain Block Matching Algorithm with numerical example.
8. Explain Motion Detection and Motion Estimation Models.
9. Explain KLT Tracker with architecture and applications.
10. Compare Sparse and Dense Optical Flow.

These answers are now at the **RGPV 7-mark level**. For **14-mark university answers**, you should further add **labeled diagrams, mathematical derivations of Optical Flow and Lucas-Kanade equations, solved numericals, and examiner-point keywords** to reach 4–5 pages per answer.